JobPulse 📊

ENGINEERING REPORT

Project Audit & Technical Roadmap • Author: Akshay • Target Stack: Spring Boot & MySQL

1. Executive Summary

JobPulse is a full-stack Spring Boot application built to organize and streamline the software engineering job search. It provides candidates with a unified portal to track company applications, compensation, interview round statuses, and recruitment pipelines.

The project currently implements an **N-Tier Architecture** (Controller → Service → Repository → MySQL Database) with JPA/Hibernate, auto-updating business logic, soft deletion, and a lightweight web dashboard. This document benchmarks existing accomplishments against commercial industry platforms (e.g., Huntr, Teal, Simplify) to identify critical architectural gaps and missing features.

2. Current Accomplishments & Feature Inventory

Feature Module	Implementation Details	Status
Job Application CRUD	Manage application records: Company, Role, Location, Job URL, Priority (`HIGH` , `MEDIUM` , `LOW`), Status (`ACTIVE` , `OFFER` , `REJECTED`), Applied Date, and Notes.	✅ **Accomplished**
Auditing & Lifecycle	Automatic tracking of `createdAt` and `updatedAt` via JPA lifecycle hooks (`@PrePersist` and `@PreUpdate`).	✅ **Accomplished**
Soft Deletion	Maintains data integrity via `isDeleted` flag, avoiding destructive row drops and preserving audit histories.	✅ **Accomplished**
Interview Rounds (1:N)	Dedicated `Round` entity tracking individual interview steps (HR, Technical, Managerial), sequence order, date, notes, and round status (`PENDING` , `CLEARED` , `FAILED`).	✅ **Accomplished**
Automated Status Sync	`RoundService` recalculates parent job status: Any round `FAILED` → `REJECTED` ; All rounds `CLEARED` → `OFFER` ; Otherwise → `ACTIVE` .	✅ **Accomplished**
Database Pagination	Spring Data JPA pagination (`/api/jobs/paginated`) sorted by `createdAt` descending with standard `PaginatedResponse<T>` DTO.	✅ **Accomplished**
Centralized Error Handling	`GlobalExceptionHandler` using `@RestControllerAdvice` returns standardized JSON error envelopes with timestamps and HTTP status codes.	✅ **Accomplished**

Feature Module	Implementation Details	Status
Web Dashboard	Fast, zero-dependency HTML5/CSS/JS interface with KPI metric cards, badge-coded statuses, quick search, and an inline rounds management drawer.	✅ **Accomplished**

3. What Typical Things Are Missed (Critical Gaps)

🔐 1. Authentication & Multi-Tenancy

- **Missing Spring Security:** No authentication filter chain or CSRF protection.
- **No User Isolation:** Single-tenant database; all visitors see and edit the same records.
- **Missing JWT & OAuth2:** Needs user registration, password hashing (BCrypt), and Google/GitHub social login.
- **Row-Level Security:** Applications must link to a `user_id` foreign key.

📄 2. Document & File Attachments

- **Resume Versioning:** Job seekers cannot associate specific resume versions with applications (e.g., `Resume_Java_v2.pdf`).
- **Storage Backend:** Missing multipart file upload endpoints backed by AWS S3, Cloudflare R2, or local secure storage.
- **Offer Letters & Briefs:** No storage for assignment documents or formal offer PDFs.

💼 3. Compensation & Domain Metadata

- **Salary Tracking:** Base pay, bonus, stock/equity, and expected vs. offered ranges.
- **Work Arrangement:** `REMOTE`, `HYBRID`, or `ON_SITE`.
- **Application Source:** LinkedIn, Referral, Indeed, Campus Placement.
- **Recruiter CRM:** Recruiter contact names, emails, phone numbers, and LinkedIn URLs.

🔄 4. Granular Pipeline Stages

- Real workflows need statuses beyond 3 basic states:
- `WISHLIST` (Bookmarked to apply)
- `APPLIED` (Awaiting response)
- `OA_RECEIVED` (Online coding challenge)
- `INTERVIEWING` (Active rounds)
- `OFFER_ACCEPTED` / `OFFER_DECLINED`
- `GHOSTED` (No response for 30+ days)

⏰ 5. Notifications & Reminders

- **Interview Alerts:** Automated reminders 24h before scheduled rounds.
- **Follow-up Triggers:** Alerts for applications inactive for >10 business days.
- **Calendar Sync:** Export `.ics` files or sync directly with Google Calendar.

📈 6. Visual Analytics & Reports

- **Recruitment Funnel:** Visual conversion rates (Applied → OA → Interview → Offer).
- **Pace & Volume Charts:** Applications submitted per week/month.
- **Response Times:** Average days between application and first interview.

4. Identified Code Issues & Fixes in Current Base

> ⚠️ **Discrepancy 1: Status Not Updated on Edit**
> In `JobService.updateJob()`, all fields are updated except `status` (omitted

`existing.setStatus(updatedJob.getStatus())`). Updating an application status via API is silently dropped.

⚠️ **Discrepancy 2: Missing Controller Validation**
Although entity models contain `@NotBlank` and `@NotNull`, `JobController` methods lack `@Valid` on `@RequestBody`. Unvalidated payloads bypass validation.

⚠️ **Discrepancy 3: Client Filter vs. Server Pagination Mismatch**
`dashboard.html` filters only the local `allJobs` array (the current 10 items). If a company is on Page 2, filtering from Page 1 yields zero results. Search/filtering must be processed via backend query parameters.

5. Phased Engineering Roadmap

PHASE 1	PHASE 2	PHASE 3	PHASE 4	PHASE 5
Bug Fixes & Stability	**Auth & Multi-Tenancy**	**Domain Expansions**	**Analytics & Alerts**	**DevOps & OpenAPI**
Status update fix, @Valid, server-side search	Spring Security, JWT, User isolation	Salary, Resume uploads, Recruiter CRM	Funnel charts, Calendar sync, Follow-ups	Swagger UI, Docker Compose, Unit tests